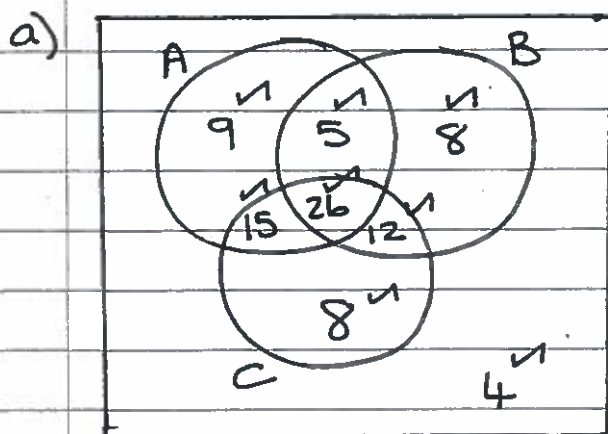


MEMO - GR 11 AP STATS

Question 1 [13]



b) $87 \checkmark \checkmark$ (2)

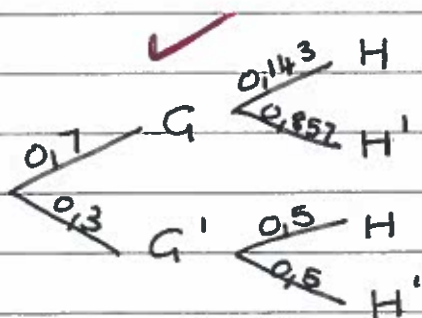
c) (i) $P(A \cup B)$
 $= \frac{75}{87} \checkmark \checkmark (0,862)$ (2)

(ii) $P(A \cap B)$
 $= \frac{31}{87} \checkmark \checkmark (0,356)$ (2)

(iii) $P(B|C)$
 $= \frac{P(B \cap C)}{P(C)} \checkmark$
 $= \frac{38/87}{61/87} \checkmark$
 $= \frac{38}{61} \checkmark (0,623)$ (3)

Question 2 [8]

Not independent



a) $P(H) \checkmark \checkmark$
 $= (0,7)(0,143) + (0,3)(0,5)$
 $= 0,2501 \checkmark$ (3)

b) $P(G \cap H) \checkmark \checkmark$
 $= 0,1001$ (2)

c) $P(G \cup H) \checkmark$
 $= P(G) + P(H) - P(G \cap H)$
 $= 0,7 + 0,2501 - 0,1001$
 $= 0,85 \checkmark \checkmark$ (3)

Question 3 [11]

a) (i) $P(A) = 0,49 \checkmark$ (1)

(ii) $P(B|A) = \frac{P(B \cap A)}{P(A)}$
 $= \frac{0,387 \checkmark}{0,49 \checkmark}$
 $= 0,79 \checkmark$ (3)

(iii) $P(A'|B) = \frac{P(A' \cap B)}{P(B)}$
 $= \frac{0,294 \checkmark}{0,681 \checkmark}$
 $= 0,432 \checkmark$ (3)

$\frac{98}{227}$

b) $P(A \cap B)$ $P(A) \times P(B)$
 $= 0,387 \checkmark$ $= 0,49 \times 0,681$
 $= 0,334 \checkmark$
 $\therefore$ not independent $\checkmark$ (4)

Question 4 [14]

a) $n! = n \times (n-1) \times (n-2) \times (n-3) \dots \checkmark \checkmark$ (2)

b) ${}^{n+2}P_3 = \frac{(n+2)! \checkmark}{(n+2-3)! \checkmark}$
 $= \frac{(n+2) \times (n+1) \times (n) \times (n-1) \times (n-2) \dots \checkmark}{(n-1) \times (n-2) \times (n-3) \dots \checkmark}$
 $= (n+2)(n+1)(n) \checkmark$
 $= n(n^2 + 3n + 2)$
 $= n^3 + 3n^2 + 2n \checkmark$ (6)

$$c) {}^n P_r = 120 {}^n C_r$$

$$\frac{n!}{(n-r)!} = 120 \cdot \frac{n!}{(n-r)! r!}$$

$$n! r! = 120 n!$$

$$r! = 120$$

$$\therefore r = 5$$

(6)

Question 5 [12]

$$a) \frac{39}{52} \times \frac{38}{51} = \frac{19}{34} \quad (0,559) \quad (3)$$

$$b) \frac{1}{52} \times \frac{12}{51} = \frac{1}{221} \quad (0,0045) \quad (3)$$

$$c) \frac{4}{52} \times \frac{1}{51} = \frac{4}{2652} = \frac{1}{663} \quad (0,00151) \quad (3)$$

$$d) \frac{13}{52} \times \frac{13}{51} = \frac{13}{204} \quad (0,0637) \quad (3)$$

Question 6 [6]

$$a) \frac{8!}{2!} = 20160 \quad (3)$$

$$b) \frac{3!}{4!} = 0,25$$

As there are only 4 digits & no repeats are allowed there can only be a 1 in 4 chance of having a 5 at the end.

(3)

Question 7 [6]

$$\frac{\binom{1}{1} \binom{12}{4} \binom{39}{5}}{\binom{52}{10}} = 0,018 \quad \checkmark \checkmark$$

Question 8 [10]

$$a) \binom{5}{3} (0,6)^3 (0,4)^2 = 0,3456 \quad \checkmark \checkmark \quad (5)$$

$$b) \frac{\binom{12}{3} \binom{8}{2}}{\binom{20}{5}} = 0,3971 \quad \checkmark \checkmark \quad (5)$$

Question 9 [12]

$$a) \binom{20}{20} (0,9)^{20} (0,1)^0 = 0,12157 \quad \checkmark \quad (0,122) \quad (5)$$

$$b) \binom{20}{19} (0,9)^{19} (0,1)^1 + \binom{20}{20} (0,9)^{20} (0,1)^0 = 0,39178 \quad \checkmark \quad (7)$$

Question 10 [7]

$$a) \binom{29}{12} = 51\,895\,935 \quad \checkmark \quad (2)$$

$$b) \frac{\binom{15}{4} \binom{14}{8}}{\binom{29}{12}} = 0,079 \quad \checkmark \quad (5)$$